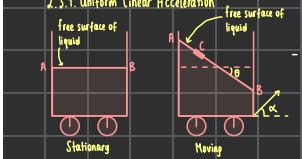
2.3. Relative Equilibrium: If a body of fluid is made to move with a constant acceleration, the liquid particles initially will move relative to each other until they start moving as a single unit-no relative motion between the liquid particles (shear stresses and shear forces between them will be zero). When the liquid comes to this static condition relative to the container, the laws of hydrostatic can be applied to determine the liquid pressure. There will arise 2 conditions:

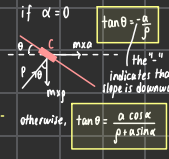
1- Liquid containers subject to constant horizontal acceleration.

2- Liquid containers subject to constant vertical acceleration.

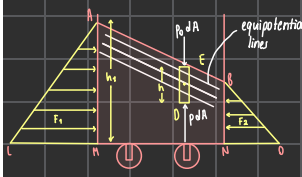
2.3.1. Uniform Linear Acceleration



if $\alpha = 0$ → $\tan\theta = \frac{a}{g}$



otherwise, $\tan\theta = \frac{a\cos\alpha}{g + a\sin\alpha}$



$p \times dA - p_0 \times dA - \rho g h dA = 0$

→ $p = p_0 + \rho g h$

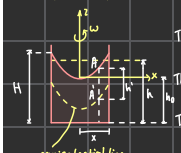
→ gauge pressure at D, $p = \rho g h$

→ for inclined acceleration,



$p = p_0 + \rho(g + a\sin\alpha)h$

2.3.2. Uniform Rotation About a Vertical Axis



The centrifugal force on a unit mass in the x-direction; $X = \frac{V^2}{x} = \frac{\omega^2 x^2}{x} = \omega^2 x$

The centrifugal force on a unit mass in the y-direction; $Y = \omega^2 y$

The gravity force on a unit-mass; $Z = -g$

→ on an equipotential line, $p = \text{constant} \rightarrow dp = 0$, $dp = \rho(X dx + Y dy + Z dz)$

$(\omega^2 x)dx + (\omega^2 y)dy + (-g)dz = 0 \rightarrow \frac{\omega^2 x^2}{2} + \frac{\omega^2 y^2}{2} - gz = C \rightarrow \frac{\omega^2 r_0^2}{2} - gz = C$

→ the pressure at a point A, $p_A = p_0 = \rho\frac{\omega^2 r_0^2}{2} - \rho g z$

→ the pressure at a point A', $p_{A'} = \rho\frac{\omega^2 r_0^2}{2} - \rho g z' + C$

→ the pressure difference between the two points, $p_{A'} - p_0 = \rho g(z - z') = \rho g h'$

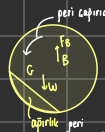
∴ in a rotating jar the pressure distribution is hydrostatic and on the bottom of the jar pressure increases from center to the edges.

→ $h_0 = h - \frac{\omega^2 r^2}{4g}$ (height prior to rotating; low point of the paraboloid)

	original	after rotation	paraboloid	(volume)

→ $\pi r^2 h = \pi r^2 H - \frac{1}{2}\pi r^2 (H - h_0) \Rightarrow 2h = H + h_0$

2.4. Stability of Floating Body: When a body is immersed in a fluid, an upward force is exerted by the fluid on the body which is equal to the weight of the fluid displaced by the body - this force is called bouyancy. The centre of bouyancy will be the center of gravity of the fluid displaced.



yer değiştiren sıvı hacminin ağırlık merkezi

G ağırlık merkezi aşağı kalkarken B noktası kayar. B noktası yeterince uzaklaşıp, bu iki kuvvet bir yeri çağırıcı moment meydana getirir.

M: metacenter (dönme öncesi ve sonrası cisme etkiyen kaldırma kuvvetlerinin etki çizgilerinin kesişimi)

MG: metacentric height

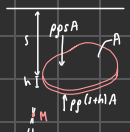
yeri çağırıcı m. ← M noktası G noktasının üstündeyse, kararlıdır (stable)

devirici m. ← M noktası G noktasının altındaysa, kararsızdır (unstable)

M noktası G noktasıyla çakışıyorsa, dengededir

GM ne kadar büyükse yüzen cisim o kadar kararlı olur.

→ $MG = \frac{I_\omega}{V} - GB$ } V: submerged volume, GB: vertical distance between the center of bouyancy and the center of gravity



$F_B = \rho g h A = \rho g V$

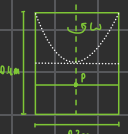


Kararlı



Kararsız

Ör: Yüksekliği 0.4m, çapı 0.2m olan düşey silindirik tanktan dörtte biri suyla yağın bir sıvıyla, kalan kısmı ise suyla doldurulmuştur. Tank düşey simetri ekseni etrafında sabit bir ω açısal hızıyla döndürülmektedir. İki sıvı arayüzünün ortasındaki P noktasının tank tabanının merkezi için geçerli açısal hızı ve bu dönme sonucu tanktan boşalacak su hacmini hesapla.



paraboloid denklemi: $z = \frac{\omega^2 r^2}{2g} + z_0$ (en alt yükseklik)

paraboloid hacmi: $V = \frac{1}{2}\pi r^2 z_p$

silindir çizimin hacmi: $V = h \times \pi \times r^2$

$z_{su} = \frac{\omega^2 r^2}{2g} + z_{alt,su} = 0.4m$

$0.4 = \frac{(13.21)^2 \times (0.15)^2}{2 \times 9.81} + z_{0,su} \Rightarrow z_{0,su} = 0.2m$

$V_{ilk} = \frac{1}{2}\pi r^2 (0.4 - 0.2) + \pi r^2 (0.2) = 0.02121$

$\pi \times r^2 \times h = \frac{1}{2}\pi \times r^2 \times z \Rightarrow 2h = z = 0.2m$

$V_{sil} = \pi r^2 h = \pi \times (0.15)^2 \times (0.4) = 0.02827$

$0.2 + \frac{\omega^2 r^2}{2g} = 0 \Rightarrow \omega = 13.21 \text{ rad/s}$

$\Delta V = 0.02827 - 0.02121 = 0.00706\ m^3$